Multiple linear regression

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Linear regression

Let's return to our key dependent and independent variable: wage and sex

```
lm(HOURPAY ~ SEX, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-16.033  -7.639  -3.384   4.256  79.396

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  22.4834     0.1959  114.76   <2e-16 ***
SEXFemale    -3.0591     0.2676  -11.43   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.44 on 7345 degrees of freedom
Multiple R-squared:  0.01748,   Adjusted R-squared:  0.01734
F-statistic: 130.6 on 1 and 7345 DF,  p-value: < 2.2e-16
```

As before, we can see the difference between the wage of men and women. Women earn around £3.06 per hour less than men. The effect is statistically significant. Sex accounts for about 1.8% of the variation in wages.

We suggested earlier that there may be other things which explain variation in wages. There may be differences between the men and women in our sample which help to explain why wages are different.

We can control for these by adding them into the model. For instance, suppose education levels vary between men and women. That may help explain why wages are different. Let's control for education in our model.

```
lm(HOURPAY ~ SEX + HIQUL22D, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-20.566  -6.337  -2.241   3.373  77.020

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                    27.1162     0.2285  118.65   <2e-16 ***
SEXFemale                      -3.6690     0.2491  -14.73   <2e-16 ***
HIQUL22DHigher education       -5.5659     0.4812  -11.57   <2e-16 ***
HIQUL22DA level or equivalent  -7.9860     0.3203  -24.93   <2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0507    0.3666  -24.69   <2e-16 ***
HIQUL22DOther qualitfication   -9.7761     0.6109  -16.00   <2e-16 ***
HIQUL22DNo Qualification      -11.7713     0.7544  -15.60   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.62 on 7340 degrees of freedom
Multiple R-squared:  0.1543,    Adjusted R-squared:  0.1536
F-statistic: 223.2 on 6 and 7340 DF,  p-value: < 2.2e-16
```

We interpret the model in much the same way as before. Here, we have two categorical variables. For each set of categorical variables, one category is excluded. We call this the reference category. In this case, the reference category for sex is male and the reference category for education is degree.

- Someone with no qualifications earns £11.77 less per hour, on average, than someone with a degree.
- A woman earns, on average, £3.67 less per hour than a man *even if they have the same level of education*

This last point is crucial. In our first model, the wage gap was £3.06. However, when we account for the differences in education, the gap widens. It seems that women have higher education levels than men, on average. Given this, we would expect them to earn more. They don't.

We are now explaining around 15% of the total variation in wages. It seems that education is important in determining wages.

Let's add a couple more variables. We can look at whether someone is on a zero hours contract (FLEX22W6) and whether they work part-time or full-time (FTPT).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-20.379  -6.206  -2.200   3.361  77.630

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                     22.0827     0.8968  24.624  < 2e-16 ***
SEXFemale                       -3.0794     0.2577 -11.949  < 2e-16 ***
HIQUL22DHigher education        -5.4010     0.4780 -11.300  < 2e-16 ***
HIQUL22DA level or equivalent   -7.7981     0.3185 -24.485  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent -8.7889     0.3652 -24.064  < 2e-16 ***
HIQUL22DOther qualitfication    -9.5050     0.6071 -15.658  < 2e-16 ***
HIQUL22DNo Qualification       -11.3429     0.7506 -15.112  < 2e-16 ***
FLEX22W6No                      5.2259     0.8751   5.972 2.46e-09 ***
FTPTPart-time                  -2.3996     0.3164  -7.584 3.77e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.54 on 7338 degrees of freedom
Multiple R-squared:  0.1668,    Adjusted R-squared:  0.1659
F-statistic: 183.6 on 8 and 7338 DF,  p-value: < 2.2e-16
```

People who are not on a zero hours contract earn £5.23 more than people who are. Someone working part-time earns, on average, £2.40 less than a full-time worker. If we look at the wage gap again, it has gone down. It is now £3.08. It seems that women are more likely to work part-time and that part-time workers are paid less than full-time workers.

Let's try controlling for length of time at current employer (EMPLEN).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN,
    data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-22.293  -6.130  -2.107   3.118  76.537

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                     21.0349     1.1507  18.281  < 2e-16 ***
SEXFemale                       -3.0489     0.2545 -11.978  < 2e-16 ***
HIQUL22DHigher education        -5.7451     0.4728 -12.151  < 2e-16 ***
HIQUL22DA level or equivalent   -7.8133     0.3146 -24.833  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0634     0.3613 -25.086  < 2e-16 ***
HIQUL22DOther qualitfication    -9.4442     0.5993 -15.758  < 2e-16 ***
HIQUL22DNo Qualification       -11.7892     0.7424 -15.880  < 2e-16 ***
FLEX22W6No                      3.8811     0.8717   4.452 8.63e-06 ***
FTPTPart-time                  -2.3085     0.3125  -7.387 1.67e-13 ***
EMPLENThree months, less than 6 -0.2034     0.9795  -0.208 0.835498
EMPLEN6 months, less than 12     0.4157     0.8990   0.462 0.643783
EMPLEN1 year, less than 2        0.1716     0.8205   0.209 0.834335
EMPLEN2 years, less than 5       1.4995     0.7837   1.913 0.055745 .
EMPLEN5 years, less than 10      2.8545     0.7971   3.581 0.000344 ***
EMPLEN10 years, less than 20     3.9389     0.7935   4.964 7.06e-07 ***
EMPLEN20 years or more           5.1767     0.8105   6.387 1.80e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.4 on 7331 degrees of freedom
Multiple R-squared:  0.189, Adjusted R-squared:  0.1874
F-statistic: 113.9 on 15 and 7331 DF,  p-value: < 2.2e-16
```

Some of the lengths of time at current employer are not statistically significant. It seems that anything less than 5 years is much the same. However, being there longer seems to provide some benefit. Our estimate of the wage gap appears to be much the same.

Let's control for whether the job is permanent or temporary (JOBTYP).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-22.291  -6.126  -2.103   3.113  76.546

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                    20.9736     1.1793  17.784  < 2e-16 ***
SEXFemale                      -3.0488     0.2546 -11.977  < 2e-16 ***
HIQUL22DHigher education       -5.7434     0.4729 -12.146  < 2e-16 ***
HIQUL22DA level or equivalent  -7.8108     0.3148 -24.810  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0590     0.3618 -25.039  < 2e-16 ***
HIQUL22DOther qualitfication   -9.4414     0.5995 -15.749  < 2e-16 ***
HIQUL22DNo Qualification      -11.7870     0.7425 -15.874  < 2e-16 ***
FLEX22W6No                     3.9228     0.8893   4.411 1.04e-05 ***
FTPTPart-time                 -2.3147     0.3136  -7.381 1.75e-13 ***
EMPLENThree months, less than 6 -0.2085     0.9798  -0.213 0.831462
EMPLEN6 months, less than 12    0.4261     0.9002   0.473 0.635947
EMPLEN1 year, less than 2       0.1803     0.8214   0.220 0.826233
EMPLEN2 years, less than 5      1.5153     0.7866   1.926 0.054088 .
EMPLEN5 years, less than 10     2.8723     0.8007   3.587 0.000336 ***
EMPLEN10 years, less than 20    3.9573     0.7973   4.963 7.09e-07 ***
EMPLEN20 years or more          5.1949     0.8142   6.380 1.87e-10 ***
JOBTYPNot permanent in some way  0.1705     0.7174   0.238 0.812127
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.4 on 7330 degrees of freedom
Multiple R-squared:  0.189, Adjusted R-squared:  0.1873
F-statistic: 106.8 on 16 and 7330 DF,  p-value: < 2.2e-16
```

It seems this is not statistically significant.

There is some evidence suggesting that the size of the firm (MPNR02) may have an effect. Let's try that.

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02, data = lfs) %>% summa
ry
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-24.110  -6.028  -2.068   3.031  77.222

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                         20.84866    1.20417  17.314  < 2e-16 ***
SEXFemale                           -3.02767    0.25204 -12.012  < 2e-16 ***
HIQUL22DHigher education            -5.41810    0.46892 -11.554  < 2e-16 ***
HIQUL22DA level or equivalent       -7.37501    0.31388 -23.496  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent     -8.49001    0.36100 -23.518  < 2e-16 ***
HIQUL22DOther qualitfication        -8.83659    0.59560 -14.836  < 2e-16 ***
HIQUL22DNo Qualification           -11.24586    0.73622 -15.275  < 2e-16 ***
FLEX22W6No                          3.32410    0.88173   3.770 0.000165 ***
FTPTPart-time                      -1.86697    0.31327  -5.960 2.65e-09 ***
EMPLENThree months, less than 6    -0.34450    0.97040  -0.355 0.722596
EMPLEN6 months, less than 12        0.33142    0.89180   0.372 0.710179
EMPLEN1 year, less than 2          -0.02982    0.81356  -0.037 0.970762
EMPLEN2 years, less than 5          1.25437    0.77936   1.609 0.107551
EMPLEN5 years, less than 10         2.45569    0.79377   3.094 0.001984 **
EMPLEN10 years, less than 20        3.43153    0.79075   4.340 1.45e-05 ***
EMPLEN20 years or more              4.54149    0.80804   5.620 1.98e-08 ***
JOBTYPNot permanent in some way    -0.06978    0.71049  -0.098 0.921763
MPNR0211-19                        -1.11462    0.52949  -2.105 0.035319 *
MPNR0220-24                        -0.71904    0.68668  -1.047 0.295076
MPNR02Don't know but under 25      -2.23238    0.89162  -2.504 0.012310 *
MPNR0225-49                        -0.72845    0.46038  -1.582 0.113631
MPNR0250-249                        0.09719    0.39421   0.247 0.805267
MPNR02250-499                       1.86456    0.56304   3.312 0.000932 ***
MPNR02Don't know but between 50 and 499   0.78142    0.65889   1.186 0.235672
MPNR02500 or more                   3.19590    0.39315   8.129 5.05e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.29 on 7322 degrees of freedom
Multiple R-squared:  0.2071,    Adjusted R-squared:  0.2045
F-statistic:  79.7 on 24 and 7322 DF,  p-value: < 2.2e-16
```

We have added a lot of categorical variables. Note that many of the levels are not statistically significant. Larger firms seem to have higher wages. We are also not explaining much additional variation.

Let's try marital status (MARSTA).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA, data = lfs)
%>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-24.798  -6.026  -1.954   3.214  76.056

Coefficients:
                                                 Estimate Std. Error
(Intercept)                                    19.919009   1.195219
SEXFemale                                      -3.006896   0.250727
HIQUL22DHigher education                       -5.492173   0.464690
HIQUL22DA level or equivalent                  -7.133012   0.311736
HIQUL22DGCSE A*-C or equivalent                -8.287454   0.358561
HIQUL22DOther qualitfication                   -9.007106   0.590139
HIQUL22DNo Qualification                      -11.161196   0.729359
FLEX22W6No                                     2.822237   0.873588
FTPTPart-time                                  -2.147829   0.311183
EMPLENThree months, less than 6                -0.385802   0.960456
EMPLEN6 months, less than 12                    0.100754   0.882826
EMPLEN1 year, less than 2                      -0.287125   0.805630
EMPLEN2 years, less than 5                      0.870865   0.771994
EMPLEN5 years, less than 10                     1.835654   0.787130
EMPLEN10 years, less than 20                    2.598980   0.785419
EMPLEN20 years or more                          3.467156   0.804347
JOBTYPNot permanent in some way                -0.085073   0.703213
MPNR0211-19                                    -1.031462   0.524134
MPNR0220-24                                    -0.625892   0.679732
MPNR02Don't know but under 25                  -2.135717   0.882739
MPNR0225-49                                    -0.670691   0.455719
MPNR0250-249                                    0.004716   0.390188
MPNR02250-499                                   1.800051   0.557272
MPNR02Don't know but between 50 and 499         0.753469   0.652287
MPNR02500 or more                               3.216213   0.389263
MARSTAMarried, living with spouse               3.173124   0.263834
MARSTAMarried separated from spouse              1.766559   0.816595
MARSTADivorced                                  3.415284   0.484361
MARSTAWidowed                                   3.524637   1.207088
MARSTACurrently or previously in civil partnership  0.154012   1.179948
                                               t value Pr(>|t|)
(Intercept)                                     16.666  < 2e-16 ***
SEXFemale                                      -11.993  < 2e-16 ***
HIQUL22DHigher education                       -11.819  < 2e-16 ***
HIQUL22DA level or equivalent                  -22.882  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                -23.113  < 2e-16 ***
HIQUL22DOther qualitfication                   -15.263  < 2e-16 ***
HIQUL22DNo Qualification                       -15.303  < 2e-16 ***
FLEX22W6No                                       3.231 0.001241 **
FTPTPart-time                                   -6.902 5.55e-12 ***
EMPLENThree months, less than 6                 -0.402 0.687927
EMPLEN6 months, less than 12                     0.114 0.909141
EMPLEN1 year, less than 2                       -0.356 0.721553
EMPLEN2 years, less than 5                       1.128 0.259326
```

```
EMPLEN5 years, less than 10                          2.332 0.019723 *
EMPLEN10 years, less than 20                         3.309 0.000941 ***
EMPLEN20 years or more                               4.311 1.65e-05 ***
JOBTYPNot permanent in some way                     -0.121 0.903712
MPNR0211-19                                         -1.968 0.049113 *
MPNR0220-24                                         -0.921 0.357189
MPNR02Don't know but under 25                       -2.419 0.015569 *
MPNR0225-49                                         -1.472 0.141139
MPNR0250-249                                         0.012 0.990357
MPNR02250-499                                        3.230 0.001243 **
MPNR02Don't know but between 50 and 499              1.155 0.248080
MPNR02500 or more                                    8.262  < 2e-16 ***
MARSTAMarried, living with spouse                   12.027  < 2e-16 ***
MARSTAMarried separated from spouse                  2.163 0.030549 *
MARSTADivorced                                       7.051 1.94e-12 ***
MARSTAWidowed                                        2.920 0.003512 **
MARSTACurrently or previously in civil partnership   0.131 0.896155
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.19 on 7317 degrees of freedom
Multiple R-squared:  0.2241,    Adjusted R-squared:  0.221
F-statistic: 72.85 on 29 and 7317 DF,  p-value: < 2.2e-16
```

Some of these levels are statistically significant. Note that our estimate of the wage gap has decreased. Now let's try some factors which ought to be very important. Let's start with industry (INDE07M).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M, da
ta = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-27.805  -5.904  -1.654   3.339  77.322

Coefficients:
                                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      18.07111    2.07800   8.696  < 2e-16 ***
SEXFemale                                        -2.12838    0.25647  -8.299  < 2e-16 ***
HIQUL22DHigher education                         -5.26106    0.45578 -11.543  < 2e-16 ***
HIQUL22DA level or equivalent                    -7.11075    0.30890 -23.020  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                  -8.19836    0.35614 -23.020  < 2e-16 ***
HIQUL22DOther qualitfication                     -8.99971    0.58280 -15.442  < 2e-16 ***
HIQUL22DNo Qualification                        -11.44694    0.71833 -15.935  < 2e-16 ***
FLEX22W6No                                       2.13716    0.85560   2.498 0.012517 *
FTPTPart-time                                    -1.30824    0.30956  -4.226 2.41e-05 ***
EMPLENThree months, less than 6                  -0.56519    0.93856  -0.602 0.547066
EMPLEN6 months, less than 12                      0.15466    0.86267   0.179 0.857720
EMPLEN1 year, less than 2                        -0.34300    0.78733  -0.436 0.663105
EMPLEN2 years, less than 5                        0.83877    0.75470   1.111 0.266434
EMPLEN5 years, less than 10                       1.79826    0.76963   2.337 0.019490 *
EMPLEN10 years, less than 20                      2.60800    0.76807   3.396 0.000689 ***
EMPLEN20 years or more                            3.69426    0.78725   4.693 2.75e-06 ***
JOBTYPNot permanent in some way                   0.51033    0.68856   0.741 0.458621
MPNR0211-19                                      -0.50335    0.51319  -0.981 0.326710
MPNR0220-24                                      -0.01248    0.66650  -0.019 0.985057
MPNR02Don't know but under 25                    -1.40105    0.86441  -1.621 0.105102
MPNR0225-49                                       0.03596    0.44944   0.080 0.936228
MPNR0250-249                                      0.49980    0.38521   1.297 0.194509
MPNR02250-499                                     1.94956    0.54714   3.563 0.000369 ***
MPNR02Don't know but between 50 and 499           1.00554    0.64016   1.571 0.116279
MPNR02500 or more                                 3.73024    0.38657   9.650  < 2e-16 ***
MARSTAMarried, living with spouse                 3.04695    0.25907  11.761  < 2e-16 ***
MARSTAMarried separated from spouse               1.84784    0.79818   2.315 0.020637 *
MARSTADivorced                                    3.52986    0.47429   7.442 1.10e-13 ***
MARSTAWidowed                                     3.68233    1.18007   3.120 0.001813 **
MARSTACurrently or previously in civil partnership  0.16740  1.15310   0.145 0.884575
INDE07MEnergy and water                           4.42746    1.91293   2.314 0.020668 *
INDE07MManufacturing                              2.28935    1.78234   1.284 0.199019
INDE07MConstruction                               4.48751    1.82011   2.466 0.013705 *
INDE07MDistribution, hotels and restaurants      -0.81363    1.76477  -0.461 0.644784
INDE07MTransport and communication                4.96552    1.77793   2.793 0.005238 **
INDE07MBanking and finance                        4.19420    1.75992   2.383 0.017189 *
INDE07MPublic admin, education and health        -0.39858    1.75291  -0.227 0.820134
INDE07MOther services                            -0.11917    1.82864  -0.065 0.948042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.952 on 7309 degrees of freedom
Multiple R-squared:  0.2602,    Adjusted R-squared:  0.2565
F-statistic: 69.48 on 37 and 7309 DF,  p-value: < 2.2e-16
```

It seems that there is some variation in pay by industry. Some male-dominated industries seem to be better paid. Note that accounting for this has reduced further our estimate of the wage gap to £2.13. This is quite a bit less than our first estimate. Let's try controlling for seniority (NSECMJ20).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + N
SECMJ20, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-29.785  -5.085  -1.209   2.884  79.650

Coefficients:
                                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      26.19639    1.93934  13.508  < 2e-16 ***
SEXFemale                                        -1.81276    0.24012  -7.550 4.90e-14 ***
HIQUL22DHigher education                         -3.17471    0.42574  -7.457 9.88e-14 ***
HIQUL22DA level or equivalent                    -4.11659    0.29875 -13.779  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                  -4.40433    0.34803 -12.655  < 2e-16 ***
HIQUL22DOther qualitfication                     -4.10139    0.56512  -7.258 4.35e-13 ***
HIQUL22DNo Qualification                         -5.92687    0.69490  -8.529  < 2e-16 ***
FLEX22W6No                                        0.39773    0.79824   0.498 0.618316
FTPTPart-time                                     0.03967    0.29256   0.136 0.892142
EMPLENThree months, less than 6                  -0.44517    0.86747  -0.513 0.607840
EMPLEN6 months, less than 12                     -0.30842    0.79761  -0.387 0.699004
EMPLEN1 year, less than 2                        -0.48591    0.72751  -0.668 0.504219
EMPLEN2 years, less than 5                        0.35393    0.69807   0.507 0.612162
EMPLEN5 years, less than 10                       0.71789    0.71281   1.007 0.313906
EMPLEN10 years, less than 20                      1.51428    0.71140   2.129 0.033322 *
EMPLEN20 years or more                            2.41304    0.72966   3.307 0.000947 ***
JOBTYPNot permanent in some way                   0.21536    0.63742   0.338 0.735473
MPNR0211-19                                      -0.14893    0.47435  -0.314 0.753551
MPNR0220-24                                       0.11269    0.61576   0.183 0.854796
MPNR02Don't know but under 25                    -1.16732    0.79908  -1.461 0.144106
MPNR0225-49                                      -0.08240    0.41601  -0.198 0.843000
MPNR0250-249                                      0.35521    0.35649   0.996 0.319086
MPNR02250-499                                     1.37499    0.50709   2.712 0.006713 **
MPNR02Don't know but between 50 and 499           0.67591    0.59225   1.141 0.253801
MPNR02500 or more                                 2.30709    0.36185   6.376 1.93e-10 ***
MARSTAMarried, living with spouse                 2.34713    0.24139   9.724  < 2e-16 ***
MARSTAMarried separated from spouse               1.98767    0.73805   2.693 0.007094 **
MARSTADivorced                                    3.18823    0.43928   7.258 4.34e-13 ***
MARSTAWidowed                                     3.32084    1.09141   3.043 0.002353 **
MARSTACurrently or previously in civil partnership  0.07321  1.06572   0.069 0.945237
INDE07MEnergy and water                           3.76222    1.76919   2.127 0.033494 *
INDE07MManufacturing                              2.05983    1.64759   1.250 0.211262
INDE07MConstruction                               3.77356    1.68292   2.242 0.024974 *
INDE07MDistribution, hotels and restaurants      -0.47190    1.63117  -0.289 0.772358
INDE07MTransport and communication                3.96325    1.64482   2.410 0.015998 *
INDE07MBanking and finance                        2.85373    1.62740   1.754 0.079550 .
INDE07MPublic admin, education and health        -0.32214    1.61998  -0.199 0.842384
INDE07MOther services                            -0.36766    1.69012  -0.218 0.827796
NSECMJ20Lower managerial and professional        -6.10448    0.29727 -20.535  < 2e-16 ***
NSECMJ20Intermediate occupations                -10.73648    0.37507 -28.625  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -10.11014    0.52727 -19.175  < 2e-16 ***
NSECMJ20Semi-routine occupations                -11.83911    0.45612 -25.956  < 2e-16 ***
NSECMJ20Routine occupations                     -12.62290    0.47731 -26.446  < 2e-16 ***
NSECMJ20Never worked                            -10.00244    0.81956 -12.205  < 2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.194 on 7303 degrees of freedom
Multiple R-squared:  0.3691,    Adjusted R-squared:  0.3654
F-statistic: 99.38 on 43 and 7303 DF,  p-value: < 2.2e-16
```

There are some very substantial differences according to level of superiority. For instance, people in routine occupations earn around £12.62 less per hour than people in higher managerial positions. This further cuts the estimate of the wage gap to £1.81.

Let's try including the AGE variable in our regression.

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + N
SECMJ20 + AGE, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + AGE, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-30.360  -5.126  -1.250   2.981  80.029

Coefficients:
                                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      2.274e+01  1.963e+00  11.584  < 2e-16 ***
SEXFemale                                       -1.753e+00  2.388e-01  -7.339 2.39e-13 ***
HIQUL22DHigher education                        -3.374e+00  4.238e-01  -7.961 1.96e-15 ***
HIQUL22DA level or equivalent                   -4.115e+00  2.970e-01 -13.856  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                 -4.799e+00  3.486e-01 -13.766  < 2e-16 ***
HIQUL22DOther qualitfication                    -4.428e+00  5.629e-01  -7.866 4.20e-15 ***
HIQUL22DNo Qualification                        -6.519e+00  6.938e-01  -9.396  < 2e-16 ***
FLEX22W6No                                       2.230e-01  7.938e-01   0.281 0.778816
FTPTPart-time                                   -9.227e-02  2.912e-01  -0.317 0.751351
EMPLENThree months, less than 6                 -4.469e-01  8.624e-01  -0.518 0.604330
EMPLEN6 months, less than 12                    -3.018e-01  7.930e-01  -0.381 0.703485
EMPLEN1 year, less than 2                       -4.897e-01  7.233e-01  -0.677 0.498380
EMPLEN2 years, less than 5                       1.750e-01  6.943e-01   0.252 0.800970
EMPLEN5 years, less than 10                      3.528e-01  7.098e-01   0.497 0.619169
EMPLEN10 years, less than 20                     8.021e-01  7.114e-01   1.127 0.259590
EMPLEN20 years or more                           1.112e+00  7.388e-01   1.505 0.132267
JOBTYPNot permanent in some way                  2.220e-01  6.337e-01   0.350 0.726105
MPNR0211-19                                     -3.772e-05  4.719e-01   0.000 0.999936
MPNR0220-24                                      1.658e-01  6.122e-01   0.271 0.786569
MPNR02Don't know but under 25                   -9.276e-01  7.949e-01  -1.167 0.243226
MPNR0225-49                                      2.462e-02  4.137e-01   0.059 0.952558
MPNR0250-249                                     5.388e-01  3.550e-01   1.518 0.129071
MPNR02250-499                                    1.496e+00  5.043e-01   2.967 0.003016 **
MPNR02Don't know but between 50 and 499          7.647e-01  5.889e-01   1.299 0.194123
MPNR02500 or more                                2.556e+00  3.607e-01   7.086 1.51e-12 ***
MARSTAMarried, living with spouse                1.446e+00  2.588e-01   5.589 2.36e-08 ***
MARSTAMarried separated from spouse              8.427e-01  7.440e-01   1.133 0.257420
MARSTADivorced                                   1.729e+00  4.640e-01   3.727 0.000195 ***
MARSTAWidowed                                    1.635e+00  1.100e+00   1.487 0.137187
MARSTACurrently or previously in civil partnership -6.873e-01  1.063e+00  -0.647 0.517819
INDE07MEnergy and water                          3.754e+00  1.759e+00   2.134 0.032857 *
INDE07MManufacturing                             2.071e+00  1.638e+00   1.264 0.206146
INDE07MConstruction                              3.781e+00  1.673e+00   2.260 0.023865 *
INDE07MDistribution, hotels and restaurants     -3.048e-01  1.622e+00  -0.188 0.850906
INDE07MTransport and communication               4.013e+00  1.635e+00   2.454 0.014148 *
INDE07MBanking and finance                       2.905e+00  1.618e+00   1.795 0.072643 .
INDE07MPublic admin, education and health       -4.107e-01  1.611e+00  -0.255 0.798724
INDE07MOther services                           -3.300e-01  1.680e+00  -0.196 0.844317
NSECMJ20Lower managerial and professional       -6.054e+00  2.956e-01 -20.479  < 2e-16 ***
NSECMJ20Intermediate occupations                -1.069e+01  3.729e-01 -28.676  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -1.001e+01  5.243e-01 -19.101  < 2e-16 ***
NSECMJ20Semi-routine occupations                -1.180e+01  4.535e-01 -26.015  < 2e-16 ***
NSECMJ20Routine occupations                     -1.275e+01  4.747e-01 -26.851  < 2e-16 ***
NSECMJ20Never worked                            -8.792e+00  8.251e-01 -10.655  < 2e-16 ***
```

```
AGE                                          1.071e-01  1.151e-02   9.306  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.14 on 7302 degrees of freedom
Multiple R-squared:  0.3765,    Adjusted R-squared:  0.3728
F-statistic: 100.2 on 44 and 7302 DF,  p-value: < 2.2e-16
```

Every year older is associated with wages being £0.11 per hour higher. Controlling for age has further reduced our estimate of the wage gap. It is now down to £1.75. Finally let's try controlling for regional variation (GOR9D).

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + N
SECMJ20 + AGE + GOR9D, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + AGE + GOR9D,
    data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-31.140  -5.101  -1.189   3.046  79.547

Coefficients:
                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                     25.78853    1.98215  13.010  < 2e-16 ***
SEXFemale                                       -1.78189    0.23677  -7.526 5.87e-14 ***
HIQUL22DHigher education                        -3.23076    0.42198  -7.656 2.16e-14 ***
HIQUL22DA level or equivalent                   -3.85940    0.29611 -13.034  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                 -4.53640    0.34719 -13.066  < 2e-16 ***
HIQUL22DOther qualitfication                    -4.44594    0.55803  -7.967 1.87e-15 ***
HIQUL22DNo Qualification                        -6.14744    0.68828  -8.932  < 2e-16 ***
FLEX22W6No                                       0.40302    0.78712   0.512 0.608648
FTPTPart-time                                   -0.07943    0.28902  -0.275 0.783459
EMPLENThree months, less than 6                 -0.39861    0.85519  -0.466 0.641152
EMPLEN6 months, less than 12                    -0.18981    0.78648  -0.241 0.809302
EMPLEN1 year, less than 2                       -0.30792    0.71735  -0.429 0.667754
EMPLEN2 years, less than 5                       0.34445    0.68859   0.500 0.616933
EMPLEN5 years, less than 10                      0.52264    0.70371   0.743 0.457692
EMPLEN10 years, less than 20                     1.06104    0.70594   1.503 0.132879
EMPLEN20 years or more                           1.57930    0.73389   2.152 0.031434 *
JOBTYPNot permanent in some way                  0.15802    0.62841   0.251 0.801461
MPNR0211-19                                      0.01830    0.46778   0.039 0.968800
MPNR0220-24                                      0.17017    0.60750   0.280 0.779394
MPNR02Don't know but under 25                   -1.01501    0.78914  -1.286 0.198407
MPNR0225-49                                      0.05431    0.41028   0.132 0.894685
MPNR0250-249                                     0.51390    0.35217   1.459 0.144539
MPNR02250-499                                    1.50815    0.50016   3.015 0.002576 **
MPNR02Don't know but between 50 and 499          0.65691    0.58493   1.123 0.261452
MPNR02500 or more                                2.40393    0.35823   6.711 2.08e-11 ***
MARSTAMarried, living with spouse                1.44619    0.25735   5.620 1.98e-08 ***
MARSTAMarried separated from spouse              0.79898    0.73767   1.083 0.278795
MARSTADivorced                                   1.68799    0.46047   3.666 0.000248 ***
MARSTAWidowed                                    1.62508    1.09137   1.489 0.136521
MARSTACurrently or previously in civil partnership -0.38073  1.05538  -0.361 0.718297
INDE07MEnergy and water                          3.70456    1.74382   2.124 0.033671 *
INDE07MManufacturing                             2.19235    1.62412   1.350 0.177099
INDE07MConstruction                              3.84059    1.65911   2.315 0.020649 *
INDE07MDistribution, hotels and restaurants     -0.39967    1.60802  -0.249 0.803718
INDE07MTransport and communication               3.64347    1.62184   2.246 0.024702 *
INDE07MBanking and finance                       2.61233    1.60432   1.628 0.103503
INDE07MPublic admin, education and health       -0.37672    1.59701  -0.236 0.813522
INDE07MOther services                           -0.61520    1.66640  -0.369 0.712006
NSECMJ20Lower managerial and professional       -5.89892    0.29352 -20.097  < 2e-16 ***
NSECMJ20Intermediate occupations               -10.52187    0.37006 -28.433  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -9.81153    0.51984 -18.874  < 2e-16 ***
NSECMJ20Semi-routine occupations               -11.54158    0.45036 -25.627  < 2e-16 ***
NSECMJ20Routine occupations                    -12.49647    0.47149 -26.504  < 2e-16 ***
```

```
NSECMJ2ONever worked                    -8.66825    0.81832 -10.593  < 2e-16 ***
AGE                                      0.10231    0.01142   8.956  < 2e-16 ***
GOR9DYorkshire and The Humber           -4.31942    0.50909  -8.485  < 2e-16 ***
GOR9DSouth West                         -3.69832    0.50952  -7.258 4.32e-13 ***
GOR9DWest Midlands                      -3.25102    0.54650  -5.949 2.83e-09 ***
GOR9DSouth East                         -2.42283    0.47010  -5.154 2.62e-07 ***
GOR9DEast of England                    -2.23360    0.53480  -4.176 2.99e-05 ***
GOR9DEast Midlands                      -4.01808    0.52771  -7.614 2.99e-14 ***
GOR9DNorth East                         -5.24307    0.65972  -7.947 2.19e-15 ***
GOR9DNorth West                         -4.00454    0.52126  -7.682 1.77e-14 ***
GOR9DWales                              -4.97962    0.66622  -7.474 8.65e-14 ***
GOR9DNorthern Ireland                   -5.08488    0.57151  -8.897  < 2e-16 ***
GOR9DScotland                           -3.28323    0.53089  -6.184 6.57e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.055 on 7291 degrees of freedom
Multiple R-squared:  0.389, Adjusted R-squared:  0.3844
F-statistic: 84.41 on 55 and 7291 DF,  p-value: < 2.2e-16
```

It seems there is regional variation. Everywhere pays less than London it seems. Note that we are talking nominal rather than real wages. That may explain some of the variation. This has once again affected our estimate of the wage gap. Now the estimate is £1.78. Now we are explaining 39% of the variation in wages. Most is still unexplained, but then our model is quite simple when you consider all the things that go into determining a person's wage.